Northeastern University

Leveraging a Convolutional Neural Network and Naive Bayesian Classifier for Acute Myeloid Leukemia (AML) Cancer Cell Classification

By: Harshini Dinesh and Samyutha Srinivasan

Purpose

- Develop a CNN model to classify AML subtypes using single-cell blood smear images.
- Build a Gaussian Naive Bayes classifier using patient-level clinical metadata for subtype prediction.
- Compare image-based (morphological) and clinical-data-based approaches for AML classification.

Background

- AML is a rapid-onset blood cancer whose complexity makes subtype identification difficult using traditional clinical methods. Accurate subtype classification is crucial to select the appropriate treatment.
- While prior work such as Madduru & Kallipalli has explored computational approaches including gradient boosting and chi-squared testing, and Yan et al. demonstrated strong predictive accuracy using SVMs and random survival forests on patient-level clinical data, few studies apply a Bayesian classifier to peripheral blood differential counts for subtype stratification.
- As image-based deep learning models, CNNs can capture morphological features in blood smear images that are not detectable through clinical data alone.
- Naive Bayes offers interpretable probabilistic outputs well-suited to the small patient cohort sizes characteristic of AML clinical datasets.

Data Overview

- Dataset: AML-Cytomorphology MLL Helmholtz dataset from The Cancer Imaging Archive.
- Includes 81,214 single-cell images from 189 patients collected between 2009–2020.
- Five classes: APL (PML::RARA), NPM1, CBFB::MYH11, RUNX1::RUNX1T1, and healthy controls.
- Images were obtained from digitized peripheral blood smears scanned at 10× and 40× magnification.
- Clinical metadata includes 13 blood differential features (e.g. leukocyte count, myeloblast percentage) plus sex and age.
- Image data and patient-level metadata correspond to the same underlying patient cohort.

Methods

Convolutional Neural Network (CNN)

A CNN model was designed to learn spatial features from single-cell images. The dataset was limited to 15,000 training images and 4,000 test images, with a patient-level 75/25 split to prevent data leakage between images from the same patient. The model uses three convolutional layers (32, 64, 128 filters) with ReLU activation and batch normalization. Global Average Pooling reduces overfitting, followed by dropout (0.4) for regularization. It is trained using the Adam optimizer with a learning rate of 1e-4 and a class-weighted cross-entropy loss to address class imbalance.

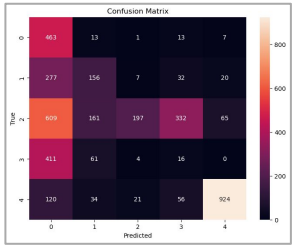
Naive Bayes

A Gaussian Naive Bayes classifier was manually implemented in R for computing the posterior probability of each AML subtype given a patient's blood counts: $P(\text{class} \mid x) \propto P(\text{class}) \times \prod P(x_i \mid \text{class})$. The numeric feature likelihoods were modeled as Gaussians parameterized by means and standard deviations c omputed for each class, with Laplace-smoothed probabilities for the categorical sex feature. The prediction was assigned to the subtype with the highest log-posterior.

Results

Convolutional Neural Network (CNN)

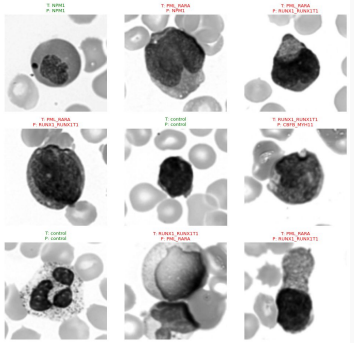
The CNN achieved an overall accuracy of 43.9%, exceeding the 20% accuracy expected from uniform random guessing in a five-class problem, demonstrating moderate predictive performance on a challenging multi-class image classification problem and significant morphological overlap of AML subtypes.



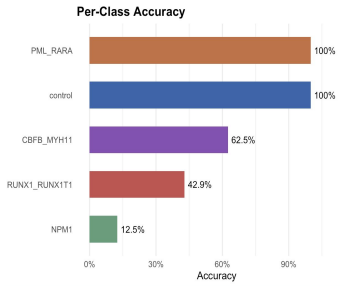
	precision	recall	f1-score	support
0	0.25	0.93	0.39	497
1	0.37	0.32	0.34	492
2	0.86	0.14	0.25	1364
3	0.04	0.03	0.03	492
4	0.91	0.80	0.85	1155
accuracy			0.44	4000
macro avg	0.48	0.45	0.37	4000
weighted avg	0.63	0.44	0.42	4000

Model shows strong bias toward predicting Class 0, with multiple misclassified samples from other classes being assigned to this category.

CNN Predictions vs. Ground Truth: A qualitative evaluation of 9 unseen test images shows both correct predictions and systematic misclassification across AML subtypes. While the model correctly identifies certain classes (e.g., control and NPM1), it frequently confuses morphologically similar subtypes such as PML-RARA and RUNX1-RUNX1T1. These patterns are consistent with the confusion matrix and classification report, reinforcing that class overlap limits performance in five-class classification.



Naive Bayes

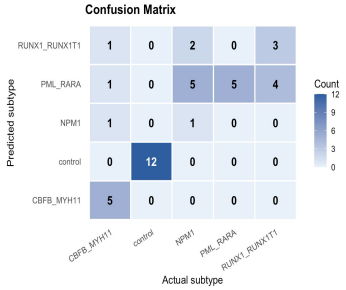


Evaluated using stratified 10-fold cross-validation (mean accuracy: 60.0%, SD: 8.2%). With a 80/20 train test split on dataset, overall accuracy was 65%.

The model achieved perfect classification for healthy controls and strong performance on PML-RARA (both 100%).

NPM1 and RUNX1-RUNX1T1 were the most difficult subtypes to differentiate.

The primary source of misclassification came from ambiguous cases from NPM1 and RUNX1-RUNX1T1 being automatically classified under PML-RARA.



Conclusions

- The CNN successfully learns meaningful morphological representations from single-cell images, as shown by performance above random chance and strong classification of at least one AML subtype. However, the confusion matrix and classification report reveal significant class overlap, particularly between morphologically similar subtypes, limiting overall predictive performance.
- Inconsistent generalization, with a gap between training and validation accuracy and unstable performance across epochs → current architecture & grayscale image representation are insufficient to fully capture all discriminative morphological patterns
- Gaussian Naive Bayes achieves 65% accuracy on AML subtype classification from peripheral blood counts alone, with perfect separation of healthy controls.
- pb_myeloblast, pb_neutrophil_segmented, and pb_monocyte are the most distinct features for AML subtype classification.

Next Steps

- Improve CNN performance using transfer learning (e.g., ResNet) and stronger data augmentation strategies to improve feature extraction and generalization.
- A Native Bayes classifier treats each blood count feature as independent given the class, ignoring biological correlations between features; future work should explore models that capture feature covariance to account for this.
- Multimodal framework: combine CNN image predictions with Naive Bayes clinical data to integrate cell morphology /blood measurements for more accurate AML classification.

References

[1] J. Al-Nusair, L. Lanino, A. Durmaz, M. G. Della Porta, A. M. Zeidan, and T. Kewan, "Artificial intelligence in myeloid malignancies: Clinical applications of machine learning in myelodysplastic syndromes and acute myeloid leukemia," Blood Reviews, vol. 74, 2025, Art. no. 101340. doi: 10.1016/j.blre.2025.101340.
[2] L. Yan, H. Yu, X. Xu, and M. Liu, "Integrated machine learning-based establishment of a prognostic model in multicenter cohorts for acute myeloid leukemia," Frontiers in Oncology, vol. 15, Art. no. 1649594, 2025, doi: 10.3389/fonc.2025.1649594.
[3] Soumya Madduru, "Impact of Machine Learning in Acute Myeloid Leukemia (AML) with Prognosis Approach for Better Accuracy", Int J Intell Syst Appl Eng, vol. 12, no. 3, pp. 3290–3295, May 2024.
[4] Hehr, M., Sadafi, A., Matek, C., Lienemann, P., Pohlkamp, C., Haferlach, T., Spiekermann, K., & Marr, C. (2023). A morphological dataset of white blood cells from patients with four different genetic AML entities and non-malignant controls (AML-Cytomorphology_MLL_Helmholtz) (Version 1) [Data set]. The Cancer Imaging Archive. https://doi.org/10.7937/6PPE-4020